Experiment-8

Write a python program to implement DFS

Aim

To write and execute a Python program to traverse a graph using Depth-First Search (DFS), which explores each branch as deep as possible before backtracking.

Algorithm

1. Represent the graph as an adjacency list.
2. Start the traversal from the start node A.
3. Mark the current node as visited and display it.
4. Visit the first unvisited neighbour of the current node using recursion.
5. If no unvisited neighbour remains, backtrack to the previous node.
6. Repeat the process until all reachable nodes are visited.
7. Display the DFS traversal order.

Coding

ex-8.py - D:/ex-8.py (3.10.0)

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```
# Depth First Search (DFS)

graph = {
    "A": ["B", "C"],
    "B": ["D", "E"],
    "C": ["F"],
    "D": [],
    "E": ["F"],
    "F": []
}

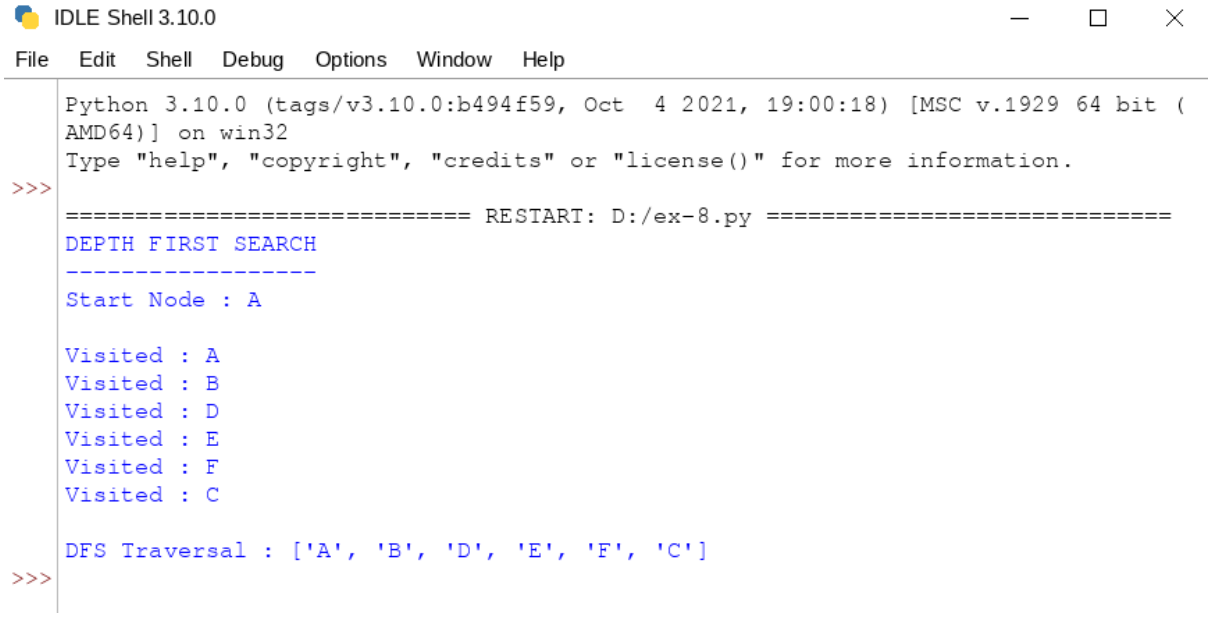
visited = []

def dfs(node):
    if node not in visited:
        visited.append(node)
        print("Visited :", node)

        # Go deeper into every unvisited neighbour
        for neighbour in graph[node]:
            dfs(neighbour)

print("DEPTH FIRST SEARCH")
print("------------------")
print("Start Node : A")
print()
dfs("A")
print()
print("DFS Traversal :", visited)
```

Output



Result

The Depth-First Search (DFS) algorithm was successfully implemented. The program explored the graph deeply along each branch and produced the order A, B, D, E, F, C.